

PEST PATROL: IMPLEMENTATION OF UAV-BASED CORN PEST MONITORING SYSTEM UTILIZING MULTIMODAL MODEL

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Abstract — Maize production in the Philippines has been increasingly affected by pest infestations, necessitating the development of advanced, scalable pest monitoring systems. This study proposes a UAV-based multimodal model for nighttime pest detection in cornfields, emphasizing its detection accuracy, confidence levels, and inference time under varying conditions. The system architecture integrates four primary components: prompt design based on pest population inputs, model integration leveraging ChatGPT-4o for inferencing, web deployment via PythonAnywhere for field accessibility, and cloud-based GPU acceleration through RunPod. Model evaluation was conducted using classification performance metrics, including precision, recall, F1-score, and confusion matrix, with pest localization as a classification criterion. Experimental trials assessed model performance across single and multiple object detection scenarios and different UAV cruising speeds. Statistical analysis using multiple one-way ANOVA revealed that pest size significantly influenced detection confidence ($p = 0.013$), while pest localization showed a marginal effect on inference time ($p = 0.080$). For deployment, pest counts were computed using a modified population density formula, yielding a root mean square error (RMSE) of 0.0781 when compared to the base formula. The results confirm the model's capability for accurate nighttime pest detection and reliable population estimation, underscoring its potential for non-intrusive pest surveillance and decision support in precision agriculture.

Keywords— Multimodal model, UAV, Nighttime Pest Monitoring, UAV, Corn Pest Detection, Pest Population Density

I. INTRODUCTION

Maize is a vital global crop, serving as a primary source of animal feed, human nutrition, and industrial raw materials [1]. In the Philippines, yellow corn is crucial to the livestock feed industry, with production targets focused on yield increases [2]. However, declining production, worsened by escalating pest infestations, particularly nocturnal moths, presents a significant challenge [3]. Key pests such as the Fall Armyworm (*Spodoptera frugiperda*), Asian Corn Borer (*Ostrinia furnacalis*), and Corn Earworm (*Helioverpa zea*) cause substantial damage through egg-laying and larval feeding [4].

Unmanned Aerial Vehicles (UAVs) have become pivotal in agricultural disease and pest detection, leveraging a range of imaging technologies and machine learning models. RGB, multispectral, hyperspectral, and thermal cameras are commonly used, with RGB sensors offering cost-effective, high-resolution imagery, while multispectral and thermal imaging provide early stress detection and physiological insights. Multispectral and thermal cameras are particularly valuable in detecting subtle stress indicators but come with higher operational costs and payload [5]. Integration of multiple sensors such as RGB, thermal, and hyperspectral, enhances detection accuracy by capturing diverse crop health indicators [6]. Studies also show that optimizing UAV flight parameters, such as low-altitude imaging and angled camera orientations, significantly improves detection of small or obscured pests [7]. These advancements have transformed monitoring from manual, labor-intensive inspections to real-time, scalable, and precise surveillance methods.

In terms of analytical techniques, machine learning and deep learning models have drastically improved detection accuracy and automation. Traditional ML methods like SVM and Random Forest rely on manually engineered features and perform well with structured datasets [8] [9] [10], while deep learning, particularly CNNs and YOLO variants, excel in image-based classification and object detection [7] [11]. Advanced models incorporating attention mechanisms, multiscale features, and semantic segmentation provide high precision, especially for small or irregularly shaped pests [7]. Multimodal approaches, combining visual data with environmental sensors or textual inputs, further improve model performance and robustness in complex field environments [12]. The emergence of large language models (LLMs) and vision-language models like CLIP enables zero-shot recognition, enhanced data annotation, and decision support [13], marking a shift toward more autonomous and intelligent pest and disease monitoring systems in agriculture.

Despite these advancements, a significant research gap exists in the use of UAVs for cost-effective, automated nighttime pest monitoring, especially in smallholder farming. While current solutions rely on expensive multispectral or

hyperspectral sensors, high-resolution RGB imagery offers a more affordable alternative for pest detection, especially in resource-constrained settings. This study aims to address this gap by proposing a UAV-based multimodal model that enhances pest detection in low-light conditions, optimizing the detection of small or partially occluded pests. By testing various operational configurations for nighttime deployment, this research seeks to provide a scalable, innovative solution for smallholder farmers, contributing to automated, intelligent crop pest monitoring systems.

The primary objective of this study is to develop and integrate a UAV-based multimodal system capable of performing pest detection during nighttime operations. The system architecture comprises four core components: prompt design based on pest population data, model integration utilizing ChatGPT-4o for inferencing, web deployment via PythonAnywhere for remote field access, and GPU-accelerated execution through RunPod. The research aims to evaluate the model's performance on key metrics in nighttime aerial pest detection, assess its capability to identify single and multiple pests within individual frames, and determine its effectiveness in estimating pest population density to support data-driven pest management strategies in cornfields.

By integrating UAV technology with multimodal AI for pest detection and density analysis, this study introduces a real-time, automated system that enhances precision agriculture. The non-invasive system offers a scalable, labor-saving alternative to manual monitoring, reducing environmental disruption, and supporting sustainable agricultural practices. The study's findings align with the UN's SDGs 12, 13, and 15, contributing to long-term goals of sustainable farming practices and precision crop management.

II. METHODOLOGY

This chapter details the researchers' methodical approach to the study, encompassing its pre-design, design, development, and testing phases, all crucial to its successful completion.

1) Concept Ideation

Corn pests progress through four stages: egg, larva, pupa, and adult (moth). The larval stage is the most damaging to corn plants, while the egg and pupal stages pose a minimal threat [11]. Detection efforts focus on adult moths, which rest on corn leaves during the day and mate at night, showing identifiable behavioral patterns [12].

Although larvae cause the most damage, detecting moths is crucial as they indicate the transition to the larval stage. Larvae emerge 2 to 7 days after hatching [13], beginning to feed corn. Monitoring moths enables early intervention before serious damage occurs.

This study uses pest population density and knowledge of corn pest behavior to develop a proactive, data-driven pest management approach. Integrating UAV technology enhances monitoring and enables timely, accurate interventions in agriculture.

2) System Design

a) System Flowchart

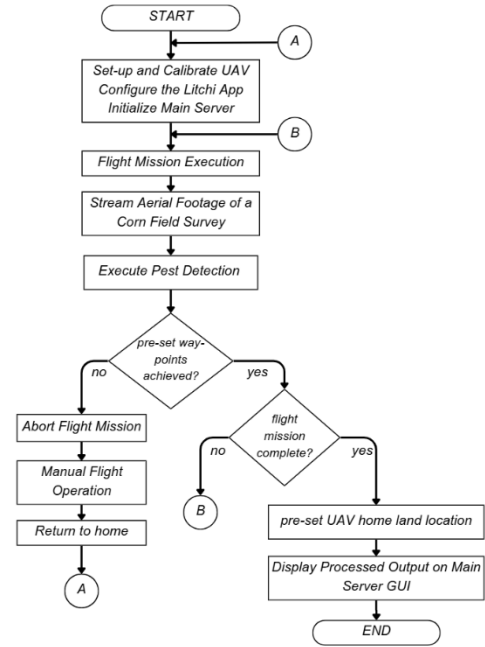


Figure 1. Automated UAV-RGB System Flowchart

The figure shows the automated UAV-RGB system for corn pest monitoring. It follows a pre-set flight plan to capture high-resolution imagery, streams data to a Multimodal Model for pest detection, and analyze pest population density.

b) System Architecture

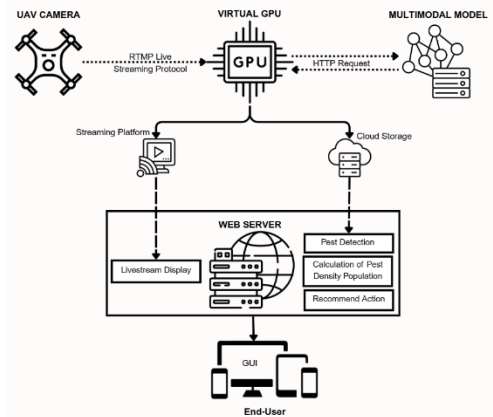


Figure 2. System Architecture of the Prototype

This figure shows a UAV pest detection system using a multimodal model. Live video is streamed via RTMP to a server, analyzed for pest detection, and VGPU used to calculate pest density. Results are shown through a desktop and mobile-friendly GUI for informed decision making.

Multimodal Model Configuration

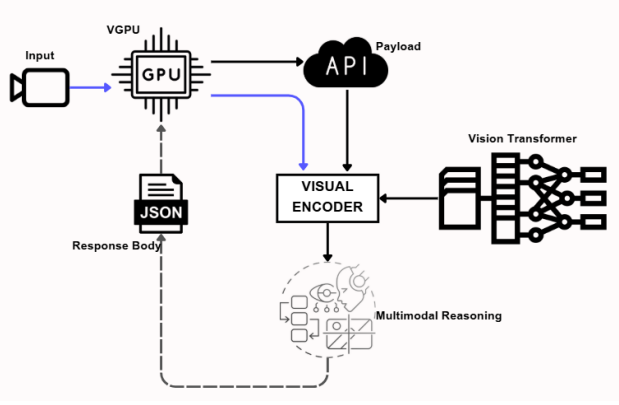


Figure 3. Multimodal Model via API Calls

The figure illustrates the internal workflow of a multimodal pest detection system. A structured JSON payload is sent via API to the model, guiding how to process encoded visual data. A Visual Encoder converts images into features for a Vision Transformer, which analyzes them in detail. These outputs feed into a multimodal reasoning module that combines visual and textual data to identify pests. Results are returned in a structured response, with accuracy heavily dependent on the quality of the initial payload, crucial for handling complex agricultural imagery.

3) Integration of System

a) Livestreaming Protocol

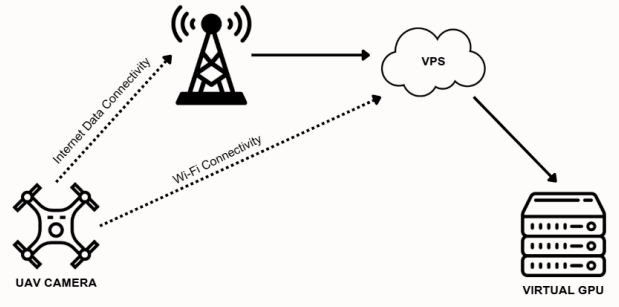


Figure 3. Livestreaming Protocol of UAV-Based Corn Pest Monitoring

The figure shows a real-time video streaming system where a UAV captures footage and transmits it to a VGPU via RTMP. The system supports both Wi-Fi and mobile data, making it suitable for remote agricultural environments with limited connectivity. NGINX, running on a VPS as an RTMP server, manages and forwards the video stream to end-users. While not optimized for high-end streaming, it provides a reliable and accessible solution for live farm monitoring.

b) GUI

The Graphical User Interface (GUI) is a key component of the pest monitoring system, offering a user-friendly layout to display pest data and recommended actions for informed decision-making.

c) Corn Pest Population Density Calculations

Pest counts detected by the AI model from UAV imagery are summed for the surveyed area. The pest population density is computed in two categories in the table below:

Modified and Base Population Density Formula

Equation 1. Population Density Base Formula:

$$D = \frac{\text{Total Pest Count } (N)}{\text{Surveyed Area } (A)} \quad (1)$$

Equation 2. Population Density Modified Formula:

$$D = \frac{\text{Total Pest Count } (N)}{\text{Surveyed Area } (A)} \times C_{ave} \quad (2)$$

where:

N = Total number of pests detected

A = Area covered during the UAV survey (in square meters)

C_{ave} = Average Confidence Level from a single detection operation

4) Testing

a) Determining the Angle of Elevation

The angle of elevation is a key factor before accuracy and deployment testing, referring to the UAV camera's vertical angle relative to the target.

The diagram illustrates the UAV camera's angle of elevation (θ) relative to the pest. Positioned above the corn tassel, this vertical angle between the camera's view and the horizontal plane impacts field of view, visibility, and detection accuracy, influencing how effectively pests are captured and identified. This configuration includes pest localization, pest size and pest orientation.

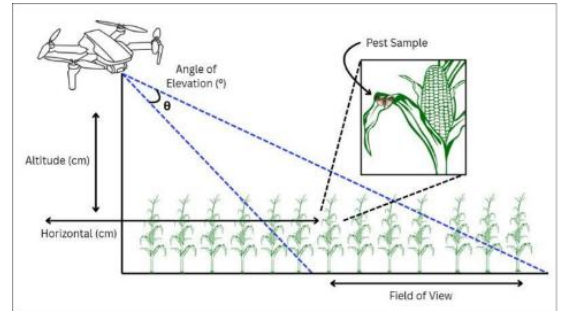


Figure 4: Illustration of Field of view and Angle of Elevation

III. RESULTS & DISCUSSION

1) Overall Setup

The image shows the full drone system setup during deployment, including the drone, main server, manual controller, a laptop running the Litchi app for flight planning, and an end-user device displaying the custom application interface.

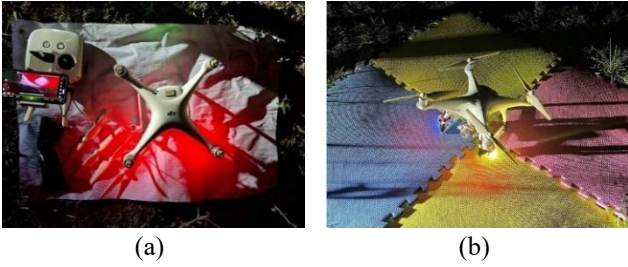


Figure 5. Deployment Setup: (a) Component Assembly, (b) Pre-Flight Setup

2) Application Software for End-Users

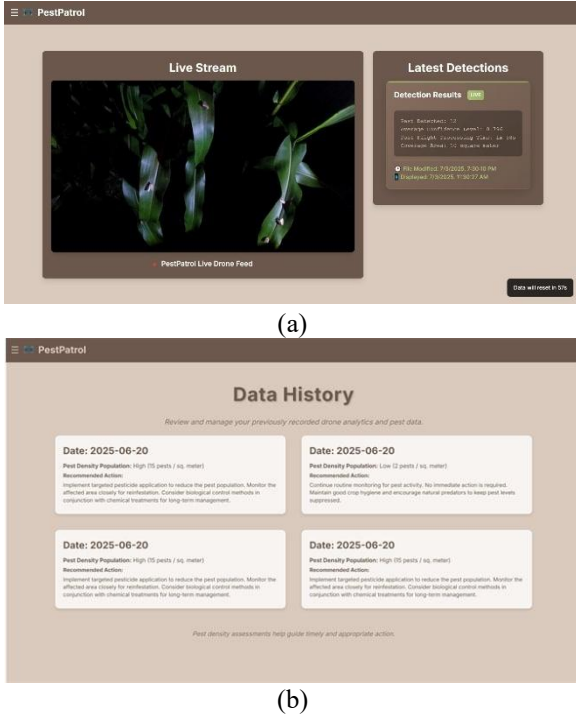


Figure 6. (a) Live Streaming Section; (b) History Section

The software supports agricultural monitoring with two main features: Live Streaming for real-time nighttime UAV views and History for pest data using a modified density formula. Together, they aid in identifying high-density zones and guide actions for precision agriculture.

3) Multimodal Model Development via API calls

The multimodal model was integrated into the UAV-based system using API calls, with an OpenAI API key enabling secure access. Python scripts running on a Virtual GPU via RunPod execute the model in real time for pest detection on a live-streaming webpage. The model processes input frames and returns detection results dynamically. A table provides an overview of the system architecture for web-based UAV pest detection using the ChatGPT-4o multimodal model.

Table 1. Development and Web-Based Integration of ChatGPT-4o for UAV Pest Detection

Component	Contribution Details	Functionality
Prompt Design	Developed a single, multi-task prompt that extracts pest count, detection confidence, inference time, and date of flight.	"12 pests detected, 92% confidence, Flight Date: 2025-07-01"

Model Integration	Integrated ChatGPT-4o to analyze UAV flight footage post-stream; system interprets pest presence from saved video, not real-time live stream.	ChatGPT-4o generates a structured report from recorded aerial data.
Web Deployment	Developed a web platform using PythonAnywhere for front-end hosting; users upload UAV video for post-flight analysis.	Users interact with the model via a web browser.
Virtual GPU Support	Employed RunPod as a virtual GPU service to ensure efficient LLM response during video analysis; connects with PythonAnywhere backend.	Enables fast and scalable inference handling behind the web interface.
Execution Mode	Used a single prompt to consolidate multiple outputs per video, improves performance and simplifies the output pipeline.	One response returns all relevant data points per UAV mission.

4) Automated UAV Flight Path Accuracy

The testing sites for drone flight missions were in Brgy. Wawa, Pililla, Rizal. These locations were selected to conduct waypoint tests in an open and controlled environment suitable for aerial navigation and data collection.

Table 2. Testing Site: Flight Path Coordinates

WP	Coordinates					
	Latitude		%Error	Longitude		%Error
	Set Value	Real Value		Set Value	Real Value	
1	14.474	14.447	0.187	121.308	121.306	0.002
2	14.474	14.488	0.097	121.308	121.32	0.01
3	14.474	14.471	0.021	121.308	121.291	0.014
4	14.474	14.494	0.139	121.308	121.307	0.002
5	14.474	14.457	0.117	121.308	121.318	0.008
6	14.474	14.476	0.014	121.308	121.322	0.012
7	14.474	14.464	0.064	121.308	121.295	0.011
8	14.474	14.477	0.022	121.308	121.324	0.013
Mean %Error			0.0826	Mean %Error		0.0088
Total %Error			0.6606	Total %Error		0.0703

The table compares the set and real coordinates for flight path A at eight waypoints, showing minimal deviation. The mean percentage error for latitude is 0.0826%, with a total of 0.6606%, while for longitude, the mean is 0.0088% and the total is 0.0703%. These results indicate high accuracy with negligible errors in both latitude and longitude.

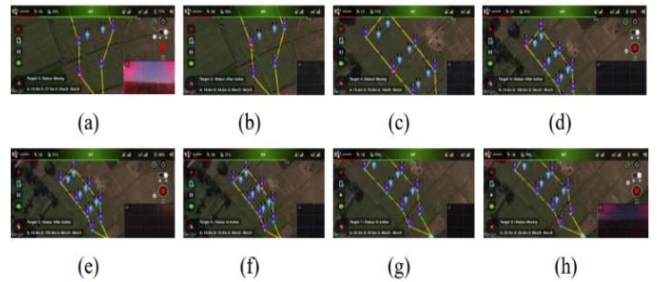


Figure 7. Flight Path Waypoints Evaluation: (a) Point 1, (b) Point 2, (c) Point 3, (d) Point 4, (e) Point 5, (f) Point 6, (g) Point 7, (h) Point 8

This flight path uses GPS to track the drone's location, showing its alignment with planned waypoints in the Litchi

interface. Accurate waypoint tracking confirms proper system operation.

The table below confirms that the drone-maintained set altitudes, demonstrating reliable flight performance. Most waypoints showed high accuracy, with the lowest at waypoint 2 (98.67%) and others ranging from 99.33% to 100%. The average accuracy is 99.65%, indicating minimal error.

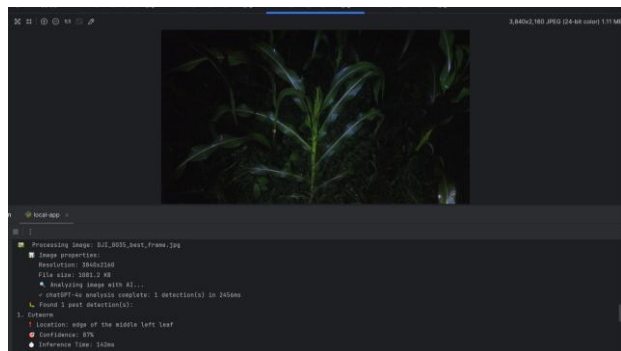
Table 3. Testing Site: Flight Path Altitude

Waypoints	Altitude (meters)		Accuracy (%)
	Set Value	Real Value	
1	15	15	100%
2	15	14.8	98.67%
3	15	15	100%
4	15	15	100%
5	20	19.9	99.5%
6	15	14.9	99.33%
7	20	20	100%
8	30	29.9	99.67%
Mean Accuracy (%)			99.65%

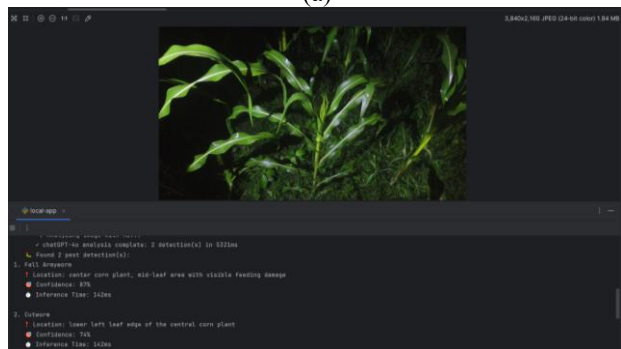
4) Performance Metrics of the UAV-Based Multimodal Model

The multimodal model was successfully integrated into the UAV mounted camera system for pest detection. Its performance was evaluated using 120 single object samples and standard classification metrics, precision, recall, F1 score, and confusion matrix. Accompanied by categorical descriptions and numerical results to assess detection accuracy and reliability.

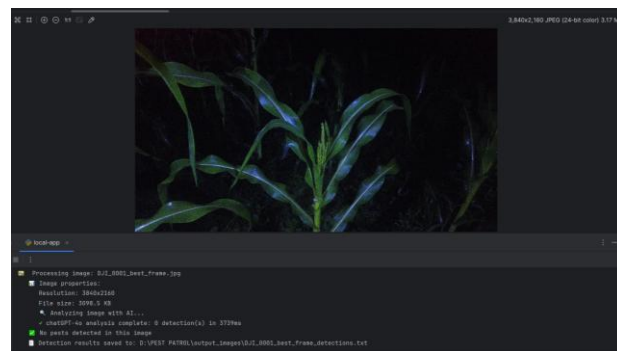
The multimodal model's performance was tested locally using UAV aerial images, with outputs categorized by confusion matrix classifications. The figure below shows sample predictions grouped by category based on the model's textual descriptions.



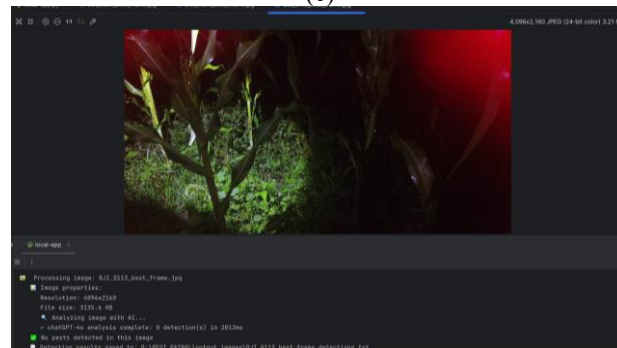
(a)



(b)



(c)



(d)

Figure 8. Sample Predictions: (a) TP, (b) FP, (c) FN, (d) TN

To support the analysis, the table below summarizes how each sample image was validated against the ground truth.

Table 4. Predication Location Vs. Ground Truth

Category	Predicted Location	Ground Truth	Remarks
True Positive	• Edge of the middle-left leaf	• Edge of the leaf	Matched
False Positive	• No pests detected center plant, mid-leaf area with visible feeding damage • Lower left leaf edge of the central corn plant	• Top of the leaf	Unmatched
False Negative	• No pests detected	• Stem of the corn plant	Unmatched
True Negative	• No pests detected	• No existing pest	Matched

The table above assesses model accuracy by comparing predicted pest locations to ground truth. A True Positive indicates a correct match (e.g., "Edge of the middle-left leaf"), while False Positives reflect incorrect or mislocated detections. False Negatives show missed pests, and True Negatives confirm accurate non-detections.

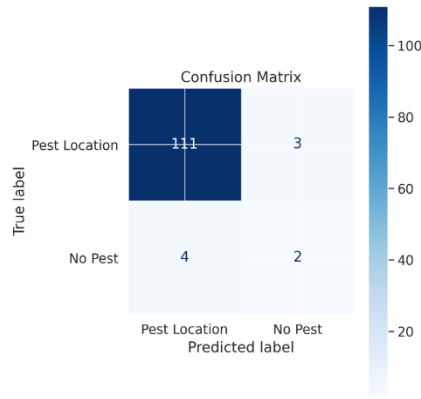


Figure 9. Confusion Matrix of the Pest Location Classification

The confusion matrix shows the model's performance: 11 True Positives, 4 False Negatives, 3 False Positives, and 2 True Negatives in classifying pest locations.

5) Evaluation of UAV-Based Multimodal Model Detection Accuracy

The multiple object detection test uses a single corn plant viewed by the UAV at 30°, 45°, and 60° angles. Starting with 2 pest samples, the number increases by 2 up to 10, using randomized configurations to evaluate the model's detection capability in a local environment.

Table 5. Testing Object Counting Accuracy under Multiple Object Conditions

Angle of Elevation (°)	Pest Sample	Pests Detected	Average Confidence Level (%)	Average Inference Time (ms)
30	2	2	86	142
	4	4	86	145
	6	6	78	160
	8	8	83	168
	10	10	79	172
Total Detection		30/30		
45	2	2	87	151
	4	4	87	149
	6	6	86	149
	8	8	84	155
	10	10	75	170
Total Detection		30/30		
60	2	2	85	160
	4	4	84	165
	6	6	84	158
	8	8	84	153
	10	10	76	173
Total Detection		30/30		

The multimodal model accurately detected all pest samples at 30°, 45°, and 60° angles, with slight variations in confidence and inference time. Confidence ranged from 75% to 87%, and inference times varied between 142ms and 173ms across angles.

6) Deployment Evaluation of the UAV Pest Monitoring System

The table summarizes 24 trials testing a modified pest density formula in a 50 m² area, with a high Mean Confidence Level (75%–88%) and more reliable performance despite variable inference times. Pest population density was

calculated using both a base and a modified formula that factors in confidence levels to account for real-world uncertainties. Over 24 trials in a 50 m² area, pest counts ranged from 19 to 20, with the base density mostly at 0.3800–0.4000 and the modified formula ranging from 0.2850 to 0.3520, reflecting consistent detection and variation in density estimates.

Table 6. Testing of a Modified Pest Density Formula over 20 Samples in a 50 m² Area

Trial	Pests Detected	Mean Confidence Level (%)	Population Density Base Formula	Modified Population Density Formula
1	20	83	0.4000	0.3320
2	20	81	0.4000	0.3240
3	19	84	0.3800	0.3192
4	20	80	0.4000	0.3200
5	19	79	0.3800	0.3002
6	20	80	0.4000	0.3200
7	20	76	0.4000	0.3040
8	19	75	0.3800	0.2850
9	20	84	0.4000	0.3360
10	20	85	0.4000	0.3400
11	20	82	0.4000	0.3280
12	20	81	0.4000	0.3240
13	19	77	0.3800	0.2926
14	20	76	0.4000	0.3040
15	19	77	0.3800	0.2926
16	19	75	0.3800	0.2850
17	20	87	0.4000	0.3480
18	20	88	0.4000	0.3520
19	20	84	0.4000	0.3360
20	20	83	0.4000	0.3320
21	20	78	0.4000	0.3120
22	20	80	0.4000	0.3200
23	19	79	0.3800	0.3002
24	19	76	0.3800	0.2888

7) Statistical Hypothesis Validation of the System

Table 7. Factor Information for Testing

Factor	Type	Levels	Values
Angle of Elevation (°)	Fixed	3	30, 45, 60
Pest Localization	Fixed	5	Back, Edge, Stem, Top, Under
Pest Orientation	Fixed	4	0, 180, 270, 90
Pest Size	Fixed	2	Large, Small
Pests Detected	Fixed	2	19, 20
Cruising Speed (m/s)	Fixed	4	0.3, 0.4, 0.5, 0.6
Gimbal Tilt Angle (°)	Fixed	3	30, 45, 60

The parameters tested under different configurations were statistically analyzed to assess the significance of each parameter's effect on confidence level and inference time in the context of object detection.

a) Multiple One-Way ANOVA for Confidence Level

The ANOVA results show that Pest Size has the most significant impact on the dependent variable, with a low P-value (0.013) and the highest F-value (6.43). In contrast, angle variables and other factors like Pest Localization, Orientation, Cruising Speed, and Gimbal Tilt show no significant effect, all with high P-values. Thus, Pest Size is the key driver of variation, with no evidence supporting angle of elevation as a contributing factor.

Table 8. Summary of Multiple One-Way Analysis of Variance at Confidence Level

Analysis	Source	DF	Adj SS	Adj MS	F-Value	P-Value
1. Angle (Multiple Object)	Angle (Multiple Object)	5	41.93	8.387	0.42	0.823
	Error	9	179	19.889		
	Total	14	220.93			
2. Single Object Analysis	Angle (Single Object)	2	0.61	0.303	0.01	0.991
	Pest Localization	4	42.92	10.729	0.32	0.864
	Pest Orientation	3	129.24	43.081	1.29	0.283
	Pest Size	1	215.34	215.344	6.43	0.013
	Error	100	3348.35	33.484		
	Total	110	3749.03			
3. Speed & Gimbal Analysis	Cruising Speed (m/s)	3	34.10	11.37	0.81	0.505
	Gimbal Tilt Angle (°)	3	49.20	16.40	1.17	0.350
	Error	17	238.14	14.01		
	Total	23	327.83			

b) Multiple One-Way ANOVA for Inference Time

Table 9. Summary of Multiple One-Way Analysis of Variance at Inference Time

Analysis	Source	DF	Adj SS	Adj MS	F-Value	P-Value
1. Angle (Multiple Object)	Angle (Multiple Objects)	5	41.93	8.387	0.42	0.823
	Error	9	179	19.889		
	Total	14	220.93			
2. Single Object Analysis	Angle (Single Object)	2	6.85	3.426	0.05	0.952
	Pest Localization	4	595.36	148.840	2.15	0.080
	Pest Orientation	3	238.64	79.548	1.15	0.333
	Pest Size	1	11.07	11.072	0.16	0.690
	Error	100	6917.42	69.174		
	Total	110	7761.75			
3. Speed & Gimbal Analysis	Cruising Speed (m/s)	3	34.1	11.37	0.81	0.505
	Gimbal Tilt Angle (°)	3	49.2	16.4	1.17	0.35
	Error	17	238.14	14.01		
	Total	23	327.83			

The table shows no statistically significant effect ($P < 0.05$) on inference time for any factor. Pest Localization, with a P-value of 0.080 and F-value of 2.15, is the closest significance. All other factors have higher P-values, confirming no meaningful impact on inference time.

c) Root Mean Square Error for Validation of Modified Pest Population Density

The validation of the modified pest population density formula was conducted using RMSE to determine the percentage error associated with the base population density formula. Statistical testing was performed using Minitab Software.

Table 10. Results on RMSE Statistics

Variable	Residuals2
N	24
Mean	0.006093
SE Mean	0.000427
StDev	0.002092
Minimum	0.002304
Q1	0.004228
Median	0.006368
Q3	0.007718
Maximum	0.009273
RMSE	0.0780577

The tables show a small average deviation (mean residual = 0.006093) between the modified and base pest density formulas. A low standard error (0.000427) and standard deviation (0.002092) indicate minimal variation. The RMSE of 0.0780577 reflects the average difference in predictions, highlighting the modified formula's overall accuracy.

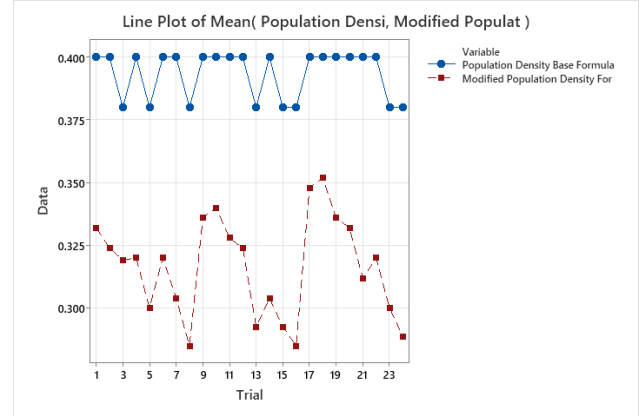


Figure 10. Line Plot of Mean between Modified and Base Formula

The line plot compares population density estimates from the base (blue, solid line) and modified (red, dashed line) formulas over 24 trials. The base formula remains relatively stable around 0.400, with slight dips at trials 3–4, 7–8, and 12–13. The modified formula fluctuates more, especially in early trials, consistently yielding lower estimates. Peaks occur around trials 10 and 17–18, and troughs around 8 and 15–16. The gap between the lines highlights the modified formula's tendency to underestimate density compared to the base.

IV. CONCLUSION

Conclusion

The RMSE value of 0.1701 and the line plot comparing the modified and base pest population density formulas indicate a small average deviation between the two, with the modified formula's estimates initially fluctuating but stabilizing over time to align more closely with the base formula. This improvement in approximation accuracy supports rejecting the null hypothesis (H_0), which claims the modified formula is not a valid approximation. Instead, the data supports the alternative hypothesis (H_a), confirming that the modified formula closely approximates the base formula with minimal deviation. These findings demonstrate the UAV-based multimodal model's effectiveness in validating pest density estimates, aiding precise and data-driven pest management in corn fields.

Recommendations

The study recommends several improvements to enhance the accuracy and efficiency of the UAV-based multimodal model for pest detection and population density estimation. These include using a high-specification RGB camera with autofocus to improve video quality and integrating a UAV-IR camera for better thermal image interpretation. Expanding the coverage area during testing can help optimize UAV flight time, while including flying

moths in the analysis broadens detection capabilities. Testing across different night-time frames and moonlight conditions will help determine optimal operating times. Additionally, performing tests from a 90-degree top-down angle can reveal discrepancies in detection performance. For improved night-time operation, an auto-adjusting flashlight synchronized with the camera is recommended, along with replacing the RGB camera with a NoIR camera for better low-light detection. Finally, incorporating AI to make decisions based on pest density and coverage area can further optimize monitoring and response strategies.

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